# LGM-Net: A *L*ightweight and *G*eneralizable Network for *M*ulti-Scale Guided Depth Super-Resolution

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***Abstract*—Guided depth map super-resolution (GDSR), as a fundamental component of 3D scene perception, aims to reconstruct high-resolution (HR) depth maps from low-resolution (LR) depth observations under the guidance of corresponding HR RGB images. However, most existing GDSR models are typically trained on a single RGB-D dataset (e.g., NYU-v2) and confined to fixed upsampling scales (e.g., $\times 4$), thereby restricting their cross-scene generalization and real-world applicability. Moreover, these methods often pursue accuracy at the expense of efficiency, leading to a sharp increase in computational cost. To comprehensively overcome these limitations, we propose *LGM-Net*, a *L*ightweight and *G*eneralizable network for *M*ulti-scale GDSR. We first design the Simple Attention and Gate mechanism with Scale-Propagation (SAGSP) block, which serves as a lightweight basic unit for efficient feature extraction with reduced parameters and memory footprint. We then introduce three effective modules: Depth-separated Cross-Modal Attention (DCMA) for bidirectional cross-modal interaction, Multimodal Dual-Attention Fusion (MDAF) for comprehensive feature integration, and Edge-Aware Dynamic Enhancement (EADE) for precise edge refinement. Building on these components, our *LGM-Net* employs a dual-branch encoder-decoder backbone with four symmetric stages to reconstruct the final high-quality depth map. To achieve high cross-scene generalization, we facilitate unified training across large-scale mixed datasets. Specifically, we adopt scale-propagation normalization along with three auxiliary loss functions that respectively constrain relative scale, gradient domain, and edge consistency. Furthermore, a training data augmentation pipeline is developed to flexibly accommodate variable upsampling scales (e.g., $\times 4$, $\times 8$, $\times 16$). Extensive experiments on public benchmarks demonstrate that our single, once-trained model *LGM-Net* achieves state-of-the-art results across all settings, and strikes an optimal balance among prediction accuracy, model capacity, and computational efficiency.**

***Index Terms*—Depth map, Super resolution, Multimodal fusion, Mixed training, Domain generalization.**

## I. Introduction

With the rapid development and widespread availability of low-cost depth sensors, acquiring depth maps from three-dimensional (3D) scenes has become increasingly accessible [1]. A depth map provides per-pixel distance information from the camera to the scene, which essentially represents the structure of 3D scenes. It has been widely utilized in various applications, such as 3D reconstruction [2], autonomous driving [3], virtual reality [4], and semantic segmentation [5].

However, a significant resolution gap persists between low-cost depth sensors and RGB cameras. It causes captured depth maps to suffer from low resolution, noise interference, and edge distortion, severely restricting their utilization in downstream tasks [6]. Because high-resolution (HR) RGB images contain rich structural details, leveraging them as guidance to recover HR depth maps from low-resolution (LR) depth observations has emerged as an effective paradigm, known as Guided Depth Super-Resolution (GDSR) [7].

Existing GDSR methods can be classified into three categories: filter-based, optimization-based, and deep learning-based [8]. Benefiting from the advancements in neural networks, deep learning-based methods have demonstrated remarkable success and become the mainstream in this field. Nevertheless, existing methods often rely on complex architecture designs to pursue superior accuracy at the expense of efficiency, leading to excessive computational cost [9]. Therefore, exploring efficient multimodal fusion paradigms has become a promising direction to alleviate computational overhead while preserving structural fidelity [10]. Beyond that, they also suffer from three major challenges:

1) **Poor scene generalization:** Most existing GDSR models are trained exclusively on NYU-v2, while a few implement on RGB-D-D for real-world testing. However, both approaches exhibit inherent limitations in samples, scenes, and distortion types, which restricts their cross-scene generalization. In the future, it is inevitable to train models across large-scale mixed datasets and explore unified generalizable frameworks to boost cross-scene generalization [11], [12].

2) **Limited scale flexibility:** The architecture of most existing GDSR models only supports a fixed upsampling scale (e.g., $\times 4$) between the input LR depth map and the HR RGB image [13]. Extending to multi-scale GDSR (e.g., $\times 4$, $\times 8$, $\times 16$) requires training separate models for different scales, which is inconvenient for practical applications.

3) **Unreliable GT annotation:** Ground-truth (GT) depth maps from real-world RGB-D datasets often contain various distortions such as noises and edge errors [14], [15]. Supervising models with unreliable GT annotation propagates errors during training, ultimately leading to degraded performance [16].

In this paper, to comprehensively overcome these challenges, we make the following efforts:

First of all, for efficient feature extraction, we design the Simple Attention and Gating with Scale-Propagation (SAGSP)

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block as the basic unit. An efficient and robust feature extraction block is crucial to balance network lightweightness and the representation capacity required for cross-scene generalization. However, existing baselines such as Restormer [17] still incur a substantial memory footprint due to redundant nonlinear activations in the gate mechanism and heavy matrix multiplications in channel-dimension attention [17]. Furthermore, the conventional layer normalization is inherently unsuitable for handling the drastic scale variations across datasets. We systematically eliminate redundant activations and projections in conventional gating and attention mechanisms. In addition, we adopt the scale-propagation normalization [18] to propagate scene scale through the network. The SAGSP block elegantly integrates these components to reduce computational complexity and parameter count while preserving global information aggregation, channel interaction, and nonlinear representation capabilities [19].

To bridge the modality gap between the RGB guidance and depth map, we then design three key modules. First, the Depth-separated Cross-Modal Attention (DCMA) promotes bidirectional interaction by transferring geometric properties while suppressing texture interference. Second, the Multi-modal Dual-Attention Fusion (MDAF) achieves cross-modal integration by adaptively recalibrating interacted features via parallel attentions. Third, the Edge-Aware Dynamic Enhancement (EADE) module dynamically refines fine-grained structural details by formulating a spatial weight mask from Laplacian edge priors and multi-scale semantic differences.

By integrating the aforementioned modules, we construct *LGM-Net*, a ***L***ightweight and ***G***eneralizable network for ***M***ulti-scale GDSR. It employs a dual-branch encoder-decoder backbone to process the HR RGB guidance and bicubic-upsampled LR depth map. Across four symmetric stages, SAGSP groups efficiently extract multi-scale features, while DCMA modules facilitate bidirectional cross-modal interactions. Aided by skip connections for information reuse, the decoder progressively restores spatial resolutions. Ultimately, the MDAF module comprehensively fuses the dual-modal representations, and the EADE module dynamically refines edge details to reconstruct the final HR depth map.

To improve scene generalization of multi-scale GDSR, we further facilitate unified training on large-scale mixed datasets by introducing scale-propagation normalization and three domain-specific loss functions. On the one hand, scale-propagation normalization preserves fast convergence and propagates scale information across diverse scenes (e.g., indoors *vs.* outdoors). On the other hand, we employ a standardized loss to align statistical distributions, a multi-scale gradient loss to constrain structural patterns, and an edge consistency loss to supervise depth predictions with reliable RGB boundaries, preventing overfitting to inherent GT distortions [14].

To enhance the scale flexibility of multi-scale GDSR, we additionally develop a training data augmentation pipeline. In detail, we perform stochastic multi-scale downsampling-upsampling cascades to simulate real-world blur degradations. Consequently, depth super-resolution tasks at various scales are converted into a generalized image restoration framework. This strategy enables a single, once-trained model to support variable upsampling scales (e.g., $\times 4$, $\times 8$, $\times 16$), while other methods often require separate models for specific scales.

Our model is once trained on large-scale mixed RGB-D datasets and comprehensively evaluated on four unseen benchmarks across (e.g., $\times 4$, $\times 8$, $\times 16$) GDSR tasks. Our model is compared with the officially released models of fifteen state-of-the-art methods, all of which require training separate models for fixed upsampling scales. Nonetheless, our single, once-trained model LGM-Net achieves state-of-the-art results across all evaluations under both quantitative and visual comparisons. Ultimately, our LGM-Net strikes an optimal balance among prediction accuracy, model capacity, and computational efficiency.

Our main contributions are summarized as follows:

- We propose a *lightweight* network for multi-scale GDSR, which employs the SAGSP block for efficient feature extraction and three key modules for effective interaction, cross-modal fusion, and precise structure refinement.
- We achieve cross-scene *generalization* for multi-scale GDSR with a robust regularization strategy, which enables unified learning across mixed datasets and prevents network overfitting to inherent GT depth distortions.
- We enhance the model *flexibility* for multi-scale GDSR to accommodate variable upsampling scales (e.g., $\times 4$, $\times 8$, $\times 16$), which employs a data augmentation pipeline to simulate various real-world blur degradations of depth data.
- Our single, once-trained model *LGM-Net* strikes an optimal balance among prediction accuracy, model capacity, and computational efficiency, when compared to state-of-the-art methods that trained separate models for fixed scales.

## II. Related Work

### A. Filter-based Methods

The key concept of filter-based methods is to design specific filter kernels to weigh spatial neighborhood pixels, thereby propagating structural priors from color guidance images to depth modalities. Over the past decades, classic approaches have utilized various manually engineered kernels, such as joint bilateral upsampling [20], non-sharpening mask guided filtering [21], and static-dynamic filters [22]. While these methods successfully transfer basic structural information, these hand-crafted designs typically extract only shallow cross-modality features. Consequently, when explicit inconsistencies arise between RGB images and depth maps (e.g., shadows or redundant textures), these methods are prone to convey misleading information, resulting in structural deformations and texture-copy artifacts.

### B. Optimization-based Methods

Optimization-based approaches formulate GDSR as a global energy minimization problem and then adopt various prior knowledge for reconstruction. Typical frameworks employ hand-crafted regularization terms, such as multi-resolution Markov Random Fields [23], dense conditional random fields [24], adaptive autoregressive models [25], and Graph Laplacian [26], to preserve depth boundaries and ensure

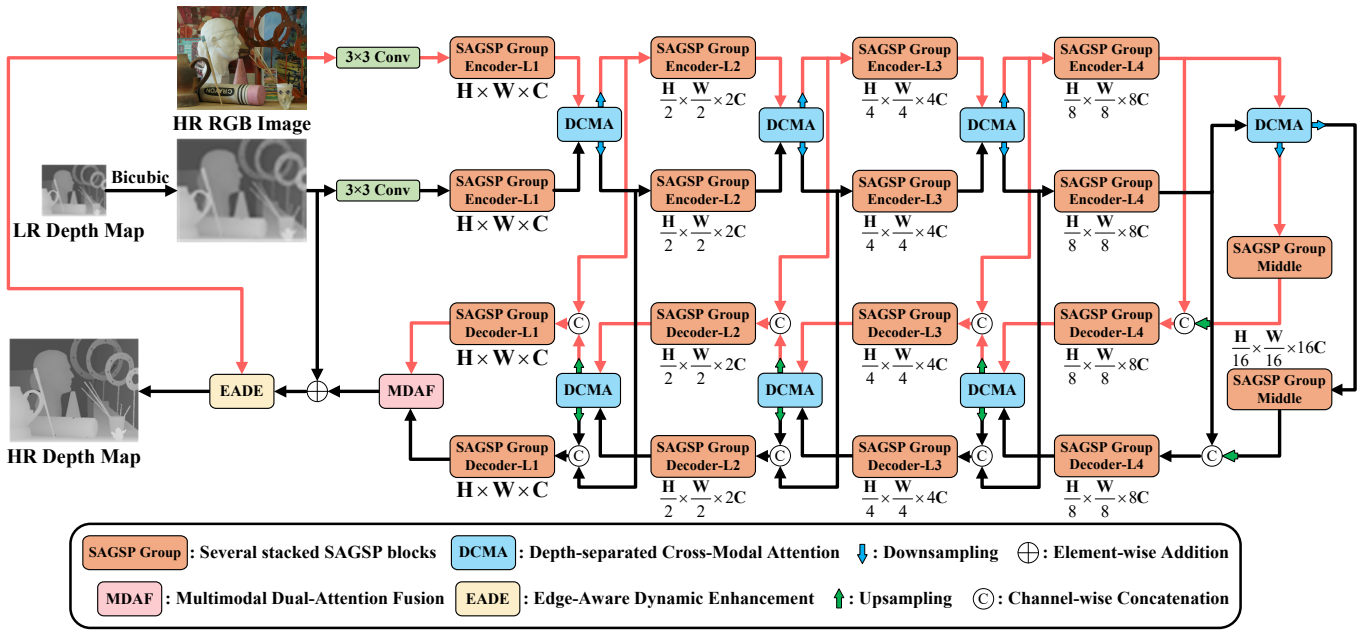



Fig. 1. Overall architecture of LGM-Net. Given an HR RGB image and an LR depth map, the latter is first upsampled via bicubic. Subsequently, a dual-branch encoder-decoder backbone extracts multi-scale features through SAGSP groups. At each stage, DCMA facilitates bidirectional interaction between RGB and depth. Finally, the cross-modal representations are integrated by MDAF, and EADE refines the structural boundaries to generate the predicted HR depth map.

smoothness. Although optimization-based methods have made notable progress in constraining global consistency, their reliance on manually designed objective functions fundamentally limits their representational capacity and adaptability in complex, real-world scenes.

### C. Deep Learning-based Methods

Driven by the rapid breakthroughs in deep neural networks, deep learning-based GDSR methodologies have achieved remarkable success by establishing end-to-end nonlinear mappings from LR depth inputs to HR depth outputs. Recent works primarily focus on designing cross-modality feature fusion modules, exploring multi-modal attention [27], cross-task knowledge distillation [28], neural implicit image functions [29], and cascaded Transformers [30].

Despite significant performance improvements, these methods prioritize increasingly complex modules, inevitably leading to a sharp rise in computational complexity that hinders edge deployment. Furthermore, most existing models face two critical bottlenecks: data overfitting and scale inflexibility. First, they are predominantly overfitted to specific datasets (e.g., NYU-v2), and their reliance on conventional normalizations (BN/LN) with fixed input-size ratios restricts them from handling drastic scale variations and diverse degradations.

To overcome these limitations, our LGM-Net conceptualizes GDSR as a generalized image restoration framework. Driven by the efficient architectural backbone with a generalizable mixed training strategy, this paradigm transforms multi-scale GDSR tasks into a unified deblurring-like image restoration task, and empowers a single, once-trained model to complete robust, multi-scale GDSR tasks with lower computational cost.

## III. Method

In this section, we introduce our LGM-Net. First, we present the problem formulation of GDSR in Section III-A. Next, we describe the overall architecture of LGM-Net in Section III-B, followed by detailing its key modules (i.e., the SAGSP block, DCMA, MDAF, and EADE) in Section III-C. Finally, we outline the loss functions and data augmentation pipeline in Section III-D and Section III-E, respectively.

### A. Problem Formulation

The goal of GDSR is to recover an HR depth map $D_{hr} \in \mathbb{R}^{1\times sH\times sW}$ from the LR depth map $D_{lr} \in \mathbb{R}^{1\times H\times W}$ under the guidance of the corresponding HR RGB image $I_{hr} \in \mathbb{R}^{3\times sH\times sW}$, where $H$, $W$ and $s$ represent the height, width of $D_{lr}$, and the scaling factor of super-resolution, respectively. Specifically, we feed the paired $D_{lr}$ and $I_{hr}$ into our LGM-Net. During training, the network gradually learns a nonlinear mapping between the input $D_{lr}$ and ideal output $D_{gt}$. Finally, the predicted HR depth map can be formulated as follows:

$$D_{hr} = \mathcal{F}(D_{lr}, I_{hr}, \theta) + f_{bicubic}(D_{lr}), \tag{1}$$

where $\mathcal{F}$ represents the nonlinear mapping function implemented by our LGM-Net, $f_{bicubic}$ denotes the bicubic interpolation operator, and $\theta$ refers to learnable network parameters.

### B. Network Architecture

As illustrated in Fig. 1, the overall architecture of our LGM-Net consists of a dual-branch encoder-decoder backbone with several critical modules. Specifically, the network takes an HR RGB guidance image $I_{hr}$ and an LR depth map $D_{lr}$ as inputs. First, $D_{lr}$ is upsampled via bicubic interpolation to generate a coarse HR depth map $\hat{D}_{hr}$. Then, both $I_{hr}$ and

$D_{lr}$ undergo initial feature extraction through two independent $3 \times 3$ convolutional layers.

Following preliminary extraction, the features are fed into a four-stage encoder to process the RGB and depth features separately. Within each encoding stage, multiple stacked SAGSP blocks form a SAGSP group to perform efficient feature extraction, followed by the DCMA module that facilitates bidirectional cross-modal interaction between RGB and depth. Then a $2 \times 2$ convolution layer with stride 2 is applied for downsampling, which halves spatial resolution while doubling the channel dimensions.

Subsequently, the high-level semantic features from the encoder are transferred to a symmetric four-stage decoder via a middle module. During the decoding process, pixel shuffling operations progressively restore the spatial resolution. Concurrently, skip connections are employed to integrate encoder features of corresponding resolutions, promoting multi-scale information propagation and reuse. Then, the RGB and depth features outputted by the decoder achieve comprehensive fusion through the MDAF module.

Finally, the EADE module dynamically refine the structural boundaries and edge details of the predicted HR depth map.

Through effective collaboration of efficient extraction, bidirectional interaction, comprehensive fusion, and dynamic edge enhancement, LGM-Net significantly improves the structural details of the predicted depth maps. Meanwhile, it strictly preserves geometric consistency to achieve effective GDSR.

### C. Key Modules

In this section, we provide detailed descriptions of the core components in our LGM-Net, including the basic SAGSP block and three critical modules: DCMA, MDAF, and EADE.

*1) SAGSP Block:* Although the Restormer [17] provides an efficient baseline for image restoration, which is well-suited for our deblurring-like GDSR task. It still incurs substantial memory footprint due to redundant nonlinear activations in the gate mechanism and heavy matrix multiplications in channel-dimension attention. Besides, the default Layer Normalization (LN) is unsuitable for handling drastic scale variations over mixed RGB-D datasets.

To address these deficiencies, we propose a lightweight and tailored SAGSP block. As shown in Fig. 2, it introduces three critical improvements beyond the original Restormer block:

**Normalization Way:** To maintain scale consistency across diverse scenes (e.g., indoors *vs.* outdoors), we replace LN with Scale-Propagation Normalization (SPN) [18], which preserves fast convergence property as well as ensures unified cross-dataset training:

$$\mathrm{SPN}(x) = \left( \sum_{i=1}^{n} w_i \widehat{x}_i + b_i \right) x, \tag{2}$$

where $x$ represents the input features, $n$ denotes the number of channels, and $w_i, b_i$ are learnable parameters.

**Gate Mechanism:** The gate mechanism (GM) [31] is defined as:

$$\mathrm{GM}(X) = f(X) \odot \sigma(g(X)), \tag{3}$$

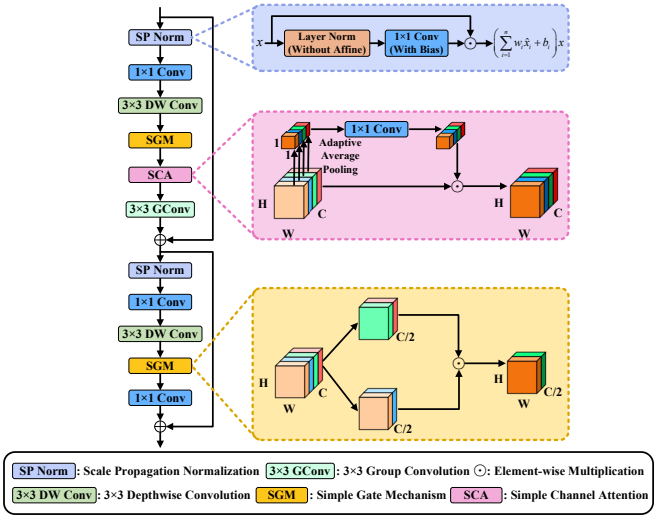



Fig. 2. Architecture of our SAGSP block. It adopts a lightweight and tailored structure for efficient feature extraction.

where $f$ and $g$ represent linear transformations, $\sigma$ denotes nonlinear activation function, and $\odot$ indicates element-wise multiplication. Meanwhile, the Restormer employs GELU [32] as gate mechanism:

$$\mathrm{GELU}(X) = X \odot \Phi(X). \tag{4}$$

The element-wise multiplication operator $\odot$ inherently introduces nonlinearity, which motivates us to remove nonlinear activation $\Phi(\cdot)$ from GELU, deriving Simple Gate Mechanism (SGM):

$$\mathrm{SGM}(X) = X_1 \odot X_2, \tag{5}$$

where $X_1$ and $X_2$ are the halves of $X$ split along the channel dimension. SGM exhibits a more streamlined structure with reduced parameters while retaining the most essential function of the gate mechanism: enhancing the nonlinear representation capability of the network.

**Channel Attention:** The Multi-Dconv Head Transposed Attention (MDTA) [17] in the Restormer block performs cross-covariance matrix multiplication, resulting in excessive memory cost. To develop a simplified attention mechanism exclusively focused on efficient feature extraction, we revisit traditional Channel Attention (CA) [33]:

$$\mathrm{CA}(X) = X * \sigma(W_{1\times1}(\delta(W_{1\times1}(\mathrm{GAP}(X))))), \tag{6}$$

where GAP represents global average pooling, $W_{1\times1}$ denotes $1 \times 1$ convolution, $\delta$ and $\sigma$ are nonlinear activation functions, and $*$ indicates element-wise multiplication with spatial broadcasting.

Similar to the SGM, we eliminate redundant activations and projections to obtain Simple Channel Attention (SCA):

$$\mathrm{SCA}(X) = X * W_{1\times1}(\mathrm{GAP}(X)). \tag{7}$$

SCA features a simplified structure with lower complexity while preserving two critical roles of channel attention: global information aggregation and channel information interaction.

*2) DCMA Module:* GDSR requires effective mechanisms to selectively transfer consistent patterns (e.g., edge details) from the RGB guidance and suppress inconsistent interference (e.g., surface textures). To this end, we design the Depth-separated Cross-Modal Attention (DCMA) to promote bidirectional interaction between RGB and depth features. As presented in Fig. 3(a), built on Scaled Dot-Product Attention (Query-Key-Value) [34], DCMA takes the intermediate dual-modality features $F_R, F_D \in \mathbb{R}^{H\times W\times C}$ as inputs.

To extract semantically aligned representations as well as capture both similar structures and distinct characteristics, we generate query matrices using SPN followed by $3\times 3$ depth-wise convolutions with shared weights $DW^S_{3\times3}$ and private weights $DW^R_{3\times3}$, $DW^D_{3\times3}$:

$$Q_R = DW^S_{3\times3}(\text{SPN}(F_R)) + DW^R_{3\times3}(\text{SPN}(F_R)), \quad (8)$$

$$Q_D = DW^S_{3\times3}(\text{SPN}(F_D)) + DW^D_{3\times3}(\text{SPN}(F_D)). \quad (9)$$

To enhance structural similarities while suppressing texture-copy artifacts during interaction, we enforce a selective constraint by setting key matrices identical to their respective query matrices. Concurrently, the raw input features are directly adopted as value matrices to maximally preserve original fidelity during propagation:

$$K_R = Q_R, \quad K_D = Q_D; \quad V_R = F_R, \quad V_D = F_D. \quad (10)$$

With the formulated queries, keys, and values, the bidirectional interacted feature maps $F_{R\to D}$ and $F_{D\to R}$ are computed as follows:

$$\begin{aligned} F_{R\to D} &= \text{Softmax}\left(\frac{Q_D K_R^T}{\sqrt{C}}\right) V_R, \\ F_{D\to R} &= \text{Softmax}\left(\frac{Q_R K_D^T}{\sqrt{C}}\right) V_D. \end{aligned} \quad (11)$$

Finally, towards integrating cross-modal guidance and retaining modality-specific identities, the outputs $F'_R$ and $F'_D$ are obtained by:

$$F'_R = \alpha F_{D\to R} + F_R, \quad F'_D = \beta F_{R\to D} + F_D, \quad (12)$$

where $\alpha$ and $\beta$ are learnable parameters that balance the contributions of bidirectional interactions and original features.

*3) MDAF Module:* Simple concatenation of multi-modal features treats all channels and spatial regions equally, failing to distinguish informative structures from redundant noise. So we propose the Multimodal Dual-Attention Fusion (MDAF) for comprehensive RGB-D integration. As depicted in Fig. 3(b), MDAF employs a dual-attention framework with channel attention (CA) and spatial attention (SA) [35] to adaptively recalibrate fused representations. It takes the features $F_R, F_D \in \mathbb{R}^{H\times W\times C}$ as inputs and concatenates them along the channel dimension to get $F_{RD} = [F_R, F_D] \in \mathbb{R}^{H\times W\times 2C}$.

Subsequently, in order to extract informative dependencies, we compute the channel weight map $A_C$ for global context aggregation and spatial weight map $A_S$ for critical regions highlighting:

$$A_C = \sigma(W_{1\times1}(\text{LeakyReLU}(DW_{3\times3}(\text{GAP}(F_{RD}))))), \quad (13)$$

$$A_S = \sigma(DW_{3\times3}([\text{Max}(F_{RD}), \text{Mean}(F_{RD})])), \quad (14)$$

where $\sigma$ denotes the Sigmoid function, and Max/Mean represent channel-wise maximum/mean operations. $W_{1\times1}$ and $DW_{3\times3}$ refer to $1\times1$ convolution and $3\times3$ depth-wise convolution, respectively.

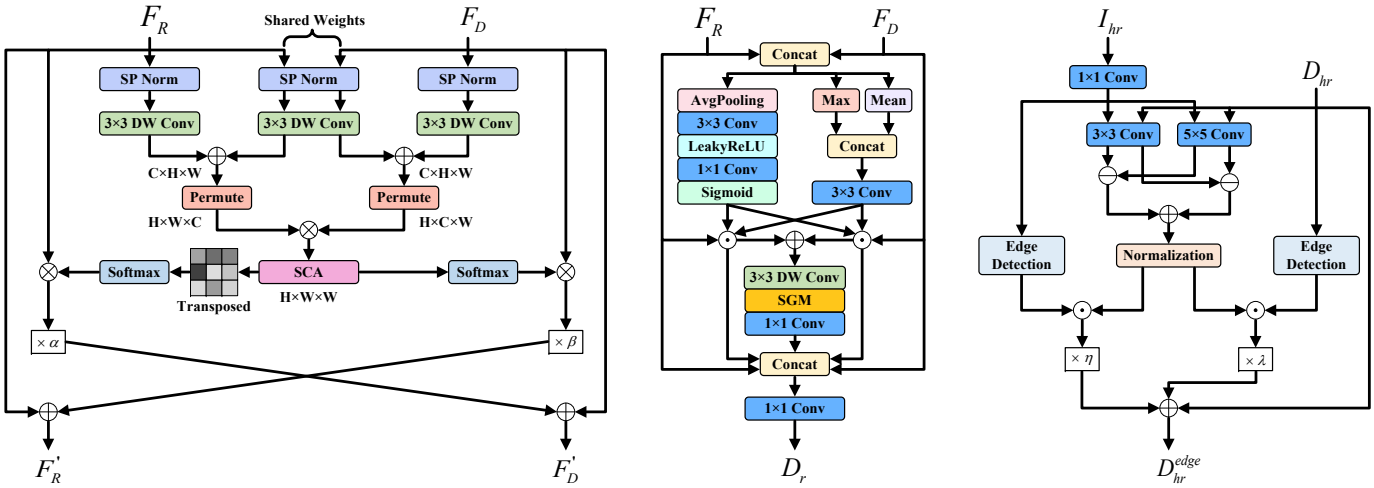
After that, the dual-attention enhanced features $F_{R_DA}$ and $F_{D_DA}$ are acquired by modulating the original features with $A_C$ and $A_S$, and we obtain the fused feature $F_{DA}$ via previously proposed SGM:

$$F_{R_DA} = F_R * A_C * A_S, \quad F_{D_DA} = F_D * A_C * A_S, \quad (15)$$

$$F_{DA} = W_{1\times1}(\text{SGM}(DW_{3\times3}(F_{R_DA} + F_{D_DA}))). \quad (16)$$

At last, to preserve multi-level representations, all middle results are concatenated and projected to get the final depth map $D_r$:

$$D_r = W_{1\times1}(\text{SAGSP}([F_R, F_D, F_{R_DA}, F_{D_DA}, F_{DA}])). \quad (17)$$




(a) DCMA Module (b) MDAF Module (c) EADE Module

Fig. 3. Architectures of the three key functional modules: (a) DCMA, (b) MDAF, and (c) EADE.

*4) EADE Module:* Although DCMA and MDAF achieve cross-modal interaction and fusion, the edge boundaries in the predicted depth map remain insufficiently sharp. To alleviate this weakness, we propose the Edge-Aware Dynamic Enhancement (EADE) to precisely preserve and refine edge details. As displayed in Fig. 3(c), EADE is built on the Laplacian edge detection and dynamic weight matrix. It takes the RGB guidance $I_{hr}$ and the predicted HR depth map $D_{hr}$ as inputs, where $D_{hr} = D_r + \hat{D}_{hr}$ is obtained by adding the MDAF output $D_r$ and the coarse HR depth map $\hat{D}_{hr}$.

First, we adopt a Laplacian kernel $K_{Laplacian}$ to explicitly extract the high-frequency edge maps from both modalities:

$$\begin{aligned} I_{edge} &= \text{Conv}(I_{hr}, K_{Laplacian}), \\ D_{edge} &= \text{Conv}(D_{hr}, K_{Laplacian}). \end{aligned} \tag{18}$$

Second, by applying convolutions with different kernel sizes (e.g., $3\times3$ and $5\times5$), we obtain multi-scale semantic features and then compute their absolute differences $\Delta I$ and $\Delta D$:

$$\begin{aligned} \Delta I &= |W_{5\times5}(I_{hr}) - W_{3\times3}(I_{hr})|, \\ \Delta D &= |W_{5\times5}(D_{hr}) - W_{3\times3}(D_{hr})|. \end{aligned} \tag{19}$$

Due to the nature of depth discontinuities, this operation yields larger values in edge regions and smaller values in flat areas. So we formulate a dynamic weight matrix $M$ for selective enhancement:

$$M = \sigma(\Delta I + \Delta D). \tag{20}$$

$M$ serves as a mask that highlights co-occurring edges between RGB and depth map, suppressing texture interference while maintaining dynamic adaptability for various inputs.

Finally, the edge-enhanced HR depth map is obtained by:

$$D_{hr}^{edge} = D_{hr} + \eta \cdot (I_{edge} * M) + \lambda \cdot (D_{edge} * M), \tag{21}$$

where $\eta$ and $\lambda$ are learnable scaling parameters.

### D. Loss Function

To achieve unified training and cross-scene generalization over mixed RGB-D datasets, and to mitigate the negative impact of inherent distortions within GT depth maps, we formulate an overall loss function that integrates four complementary components.

**Absolute-Domain Loss ($L_1$):** As the foundational supervision constraint, the standard $L_1$ loss is defined as:

$$L_1 = \frac{1}{N}\sum_{i=1}^{N} |D_{hr}^i - D_{gt}^i|, \tag{22}$$

where $N$ is the number of valid pixels of $D_{gt}$.

**Mixed-Scale Constraint Loss ($L_{std}$ and $L_{grd}$):** To align the diverse statistical distributions and scale discrepancies across different datasets, and guide the network to learn unified relative structural patterns rather than absolute magnitudes, we introduce the standardized loss $L_{std}$ and the multi-scale gradient loss $L_{grd}$:

$$L_{std} = \frac{1}{N}\sum_{i=1}^{N} \left| \frac{D_{hr}^i - \mu_{hr}}{\sigma_{hr}} - \frac{D_{gt}^i - \mu_{gt}}{\sigma_{gt}} \right|, \tag{23}$$

where $\mu_{hr}, \mu_{gt}$ and $\sigma_{hr}, \sigma_{gt}$ represent the mean and standard deviation of $D_{hr}$ and $D_{gt}$, respectively.

$$L_{grd} = \frac{1}{kN}\sum_{j=1}^{k}\sum_{i=1}^{N} \left( |\nabla_x Res_i^j| + |\nabla_y Res_i^j| \right), \tag{24}$$

where $k$ denotes the number of downsampling levels, $\nabla_x, \nabla_y$ are the Sobel operators for gradient extraction in the horizontal and vertical directions, and $Res_i = D_{hr}^i - D_{gt}^i$ refers to the residual.

**Structure Consistency Loss ($L_{esc}$):** To alleviate the negative effects of unreliable GT depth maps in real-world captured RGB-D datasets, we adopt the edge structure consistency loss $L_{esc}$ [14]. By leveraging the high-resolution and reliable edge details from corresponding guidance RGB images, this loss selectively supervises the boundaries of the predicted depth maps, thereby preventing the network from overfitting to potential GT distortions:

$$L_{esc} = \sum_{p_1, p_2 \in \Omega_{edge}} \mu_d(p_1)\rho(\omega_d(p_1), \omega_d(p_2)), \tag{25}$$

where $\Omega_{edge}$ defines the sampled edge regions, and $\mu_d \in \{0, 1\}$ acts as a mask to ensure that $L_{esc}$ exclusively constrains the edge similarity without introducing other texture artifacts. The term $\rho(\cdot,\cdot)$ is a structural similarity metric, which is evaluated based on local Gaussian similarities $\omega_d$ and $\omega_c$ within the depth and RGB domains. For mathematical details, we refer readers to the original paper.

**Total Loss ($L_{total}$):** The overall loss function $L_{total}$ is a weighted combination of all the aforementioned loss terms:

$$L_{total} = L_1 + \gamma_1 L_{std} + \gamma_2 L_{grd} + \gamma_3 L_{esc}, \tag{26}$$

where $\gamma_1$, $\gamma_2$, and $\gamma_3$ are hyperparameters for balance.

### E. Data Augmentation

The preceding architectural designs are dedicated to improving computational efficiency, cross-modal interaction, and cross-scene generalization. Beyond that, we aim to further broaden the versatility of LGM-Net, enabling a single, once-trained model to handle multi-scale GDSR tasks for enhanced convenience and adaptability.

Existing GDSR models typically operate at a fixed upsampling scale $s$, taking $D_{lr} \in \mathbb{R}^{1\times H\times W}$ and $I_{hr} \in \mathbb{R}^{3\times sH\times sW}$ as inputs. Consequently, a model trained for a specific scale (e.g., $\times4$) is not applicable to other scales (e.g., $\times8$, $\times16$). However, the training and deployment of multiple scale-specific models are computationally expensive and impractical for real-world applications.

To overcome this limitation, we propose a data augmentation pipeline that unifies multi-scale GDSR tasks into a generalized image restoration framework. As shown in Fig. 4, training data augmentation pipeline simulates blur degradations at various scales by applying stochastic distortions to GT depth maps.

Specifically, to emulate multi-scale blur degradations, we perform random multi-scale downsampling-upsampling processes (e.g., $\times4$, $\times8$, $\times16$, ... or any required scale) to the original GT depth maps. Via two inverse bicubic interpolation

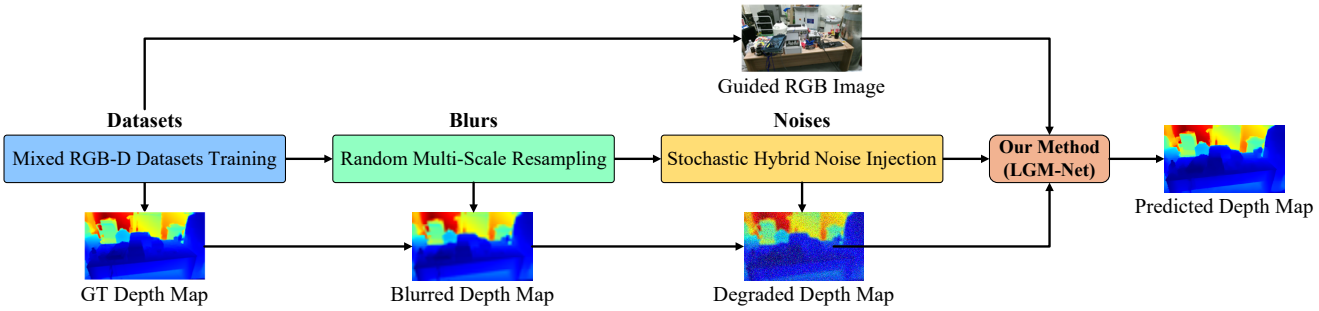



Fig. 4. Architecture of training data augmentation pipeline.

operations, this pipeline generates degraded depth maps that retain their original spatial resolutions while exhibiting blur characteristics at various scales.

Additionally, for joint depth super resolution and depth denoising tasks, we can further add random levels and types of noises to the blurred depth maps, such as Gaussian noise with random standard deviation and salt-and-pepper noise with random probability. This process produces degraded depth maps with both blurs and noises.

During training, pairs of synthetically degraded depth maps and the corresponding HR RGB images are fed into our LGM-Net. Through this process, the network is iteratively optimized to reconstruct the original GT depth maps. By exposing the model to a continuous spectrum of simulated degradation patterns, even a single model is able to robustly handle multi-scale blur distortions.

## IV. Experiments

### A. Setting

**Datasets.** Table I describes our training and testing datasets.

TABLE I
RGB-D DATASETS FOR TRAINING AND TESTING.

| Datasets | Name | Size | Scene | Sensor Type |
|---|---|---|---|---|
| Training | HRWSI | 20778 | Outdoor | Stereo Vision |
| | Matterport3D | 194400 | Indoor | Structured Light |
| | UnrealCV | 5000 | Indoor & Outdoor | Synthetic |
| | Virtual KITTI | 21260 | Outdoor | Synthetic |
| Testing | Lu | 6 | Indoor | Structured Light |
| | Middlebury | 30 | Indoor | Structured Light |
| | RGB-D-D | 405 | Indoor & Outdoor | ToF |
| | NYU-v2 | 449 | Indoor | Structured Light |

For training, we adopt large-scale mixed RGB-D datasets with various scenes and sensor types. It enables our model to learn as much knowledge as possible, thereby enhancing its generalization capability. In detail, we select four public RGB-D datasets: HRWSI (20778 pairs) [36], Matterport3D (194400 pairs) [37], UnrealCV (5000 pairs) [38], and Virtual KITTI (21260 pairs) [39]. For testing, we evaluate our model on unseen datasets to assess its cross-scene generalization across unfamiliar scenes with diverse scales and semantics. Specifically, we utilize four RGB-D benchmark datasets, including Lu (6 pairs) [40], Middlebury (30 pairs) [41], [42], RGB-D-D (405 pairs) [43], and NYU-v2 (last 449 pairs) [44].

**Metric.** To quantitatively evaluate the reconstruction accuracy, we employ the widely used Root Mean Square Error (RMSE) metric. It measures the pixel-wise discrepancy between the predicted depth map $D_{hr}$ and its corresponding ground truth depth map $D_{gt}$.

**Implementation.** For the training datasets, we randomly crop the guided RGB images and GT depth maps into patches of size $320\times320$ and uniformly normalize their values to $[0, 1]$. In the data augmentation pipeline, the standard deviation of Gaussian noise ranges from 0.01 to 0.1, the probability of salt-and-pepper noise ranges from 0 to 1, and we utilize bicubic interpolation for the downsampling-upsampling process with the scaling factor randomly selected from $\{\times4, \times8, \times16\}$. During the training process, the number of epochs and batch size are set to 300 and 16, respectively. Our model is optimized by the Adam optimizer [45] with an initial learning rate of $1 \times 10^{-4}$ and parameters set to $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The learning rate is halved every 100 epochs. We implement our model with PyTorch and conduct all training and testing processes on a single NVIDIA GeForce RTX 4090 GPU.

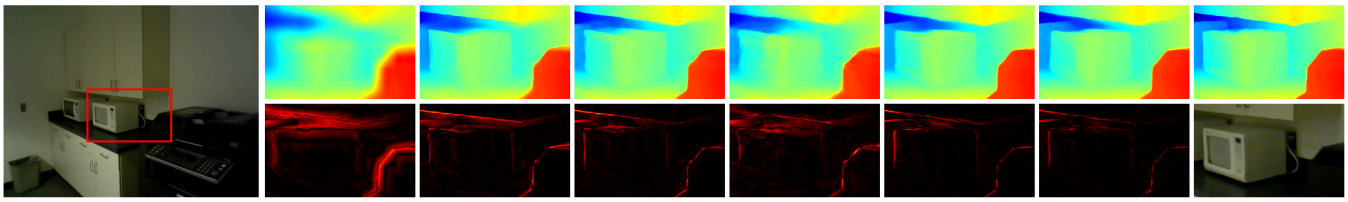
### B. Main Results

We evaluate our LGM-Net on four benchmarks across $\times4$, $\times8$, and $\times16$ GDSR tasks, and compare it with 15 state-of-the-art methods. For methods with available pre-trained models, we directly adopt them for evaluation. Otherwise, we retrain the models according to the configurations in their respective papers before testing.

**Quantitative results.** Table II summarizes the quantitative results (RMSE) of our method and 15 SOTA baselines, where lower RMSE values indicate better results. As listed in the table, our LGM-Net outperforms all existing SOTA methods, achieving the best results across most datasets and scales. This superiority is particularly evident in more challenging tasks such as $\times8$ and $\times16$ GDSR. Compared to the second-best methods, our LGM-Net reduces the average RMSE by 0.02 cm and 0.12 cm on $\times8$ and $\times16$ GDSR. It is worth noting that all benchmarks remain strictly unseen during our training phase. Furthermore, while most baselines require training separate models for scale-specific GDSR tasks (e.g., $\times4$), ours employs only a single, once-trained model to handle all multi-scale GDSR tasks (including $\times4$, $\times8$, and $\times16$). These achievements strongly validate that our method achieves both exceptional depth super-resolution accuracy and enhanced generalization capability.

TABLE II
QUANTITATIVE COMPARISONS (RMSE) WITH STATE-OF-THE-ART METHODS ON FOUR BENCHMARKS. THE BEST AND SECOND BEST VALUES ARE RESPECTIVELY HIGHLIGHTED IN **BOLD** AND UNDERLINE. NOTABLY, UNLIKE BASELINES THAT REQUIRE SEPARATE MODELS FOR EACH SPECIFIC SCALE, OUR METHOD ACCOMPLISHES MULTI-SCALE TASKS ($\times 4$, $\times 8$, AND $\times 16$) WITH ONLY A SINGLE, ONCE-TRAINED MODEL.

| Method | Lu | | | Middlebury | | | RGB-D-D | | | NYU-v2 | | | Average | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\times 4$ | $\times 8$ | $\times 16$ | $\times 4$ | $\times 8$ | $\times 16$ | $\times 4$ | $\times 8$ | $\times 16$ | $\times 4$ | $\times 8$ | $\times 16$ | $\times 4$ | $\times 8$ | $\times 16$ |
| DJF [6] | 1.65 | 3.96 | 6.75 | 1.68 | 3.24 | 5.62 | 3.41 | 5.57 | 8.15 | 2.80 | 5.33 | 9.46 | 2.39 | 4.53 | 7.50 |
| DJFR [46] | 1.15 | 3.57 | 6.77 | 1.32 | 3.19 | 5.57 | 3.35 | 5.57 | 7.99 | 2.38 | 4.94 | 9.18 | 2.05 | 4.32 | 7.38 |
| CUNet [47] | 0.91 | 2.23 | 4.99 | 1.10 | 2.17 | 4.33 | 1.18 | 1.95 | 3.45 | 1.92 | 3.70 | 6.78 | 1.28 | 2.51 | 4.89 |
| DKN [48] | 0.96 | 2.16 | 5.11 | 1.23 | 2.12 | 4.24 | 1.30 | 1.96 | 3.42 | 1.62 | 3.26 | 6.51 | 1.28 | 2.38 | 4.82 |
| FDKN [48] | 0.82 | 2.10 | 5.05 | 1.08 | 2.17 | 4.50 | 1.18 | 1.91 | 3.41 | 1.86 | 3.58 | 6.96 | 1.24 | 2.44 | 4.98 |
| FDSR [43] | 1.29 | 2.19 | 5.00 | 1.13 | 2.08 | 4.39 | 1.16 | 1.82 | 3.06 | 1.61 | 3.18 | 5.86 | 1.30 | 2.32 | 4.58 |
| JIIF [29] | 0.85 | 1.73 | 4.16 | 1.09 | 1.82 | 3.31 | 1.21 | 1.88 | 2.99 | 1.37 | 2.76 | 5.27 | 1.13 | 2.05 | 3.93 |
| DCTNet [49] | 0.88 | 1.85 | 4.39 | 1.10 | 2.05 | 4.19 | 1.08 | 1.74 | 3.05 | 1.59 | 3.16 | 5.84 | 1.16 | 2.20 | 4.37 |
| SUFT [50] | 1.10 | 1.74 | 3.92 | 1.07 | 1.75 | 3.18 | 1.10 | 1.69 | 2.71 | 1.12 | 2.51 | 4.86 | 1.10 | 1.92 | 3.67 |
| RSAG [51] | **0.79** | 1.67 | 4.30 | 1.13 | 2.74 | 3.55 | 1.14 | 1.75 | 2.96 | 1.23 | 2.51 | 5.27 | 1.07 | 2.17 | 4.02 |
| SSDNet [52] | 0.80 | 1.82 | 4.77 | 1.02 | 1.91 | 4.02 | **1.04** | 1.72 | 2.92 | 1.60 | 3.14 | 5.86 | 1.12 | 2.15 | 4.39 |
| DuCos [53] | 1.38 | 2.67 | 5.18 | 1.45 | 2.23 | 3.96 | 1.27 | 2.23 | 3.44 | 2.60 | 4.61 | 7.37 | 1.68 | 2.94 | 4.99 |
| IGAF [54] | 0.82 | 1.68 | 4.14 | **1.01** | 1.73 | 3.24 | 1.08 | 1.69 | 2.69 | 1.12 | 2.48 | 5.00 | **1.01** | 1.90 | 3.77 |
| MSF-UM [55] | 1.05 | 1.82 | 3.82 | 1.19 | 1.82 | 3.14 | 1.17 | 1.70 | 2.68 | 1.15 | 2.48 | 4.67 | 1.14 | 1.96 | 3.58 |
| SGNet [56] | 1.03 | **1.61** | 3.55 | 1.15 | 1.64 | 2.95 | 1.10 | 1.64 | 2.55 | **1.10** | 2.44 | 4.77 | 1.10 | 1.83 | 3.46 |
| **Ours** | 0.88 | 1.63 | **3.46** | 1.06 | **1.58** | **2.89** | 1.17 | **1.61** | **2.37** | 1.12 | **2.41** | **4.63** | 1.06 | **1.81** | **3.34** |



(a) RGB (b) Bicubic (c) DKN (d) JIIF (e) DCTNet (f) SGNet (g) Ours (h) GT & RGB

Fig. 5. Visual results and error maps on Lu dataset ($\times 16$).

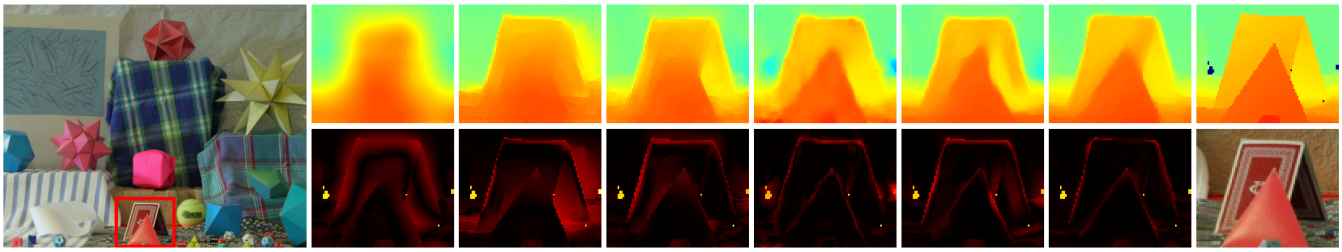
**Visual results.** In Figs. 5–7, we present the visual comparisons of our method and representative baseline methods on the $\times 16$ GDSR results. In each figure, the first row displays the predicted HR depth maps, and the second row visualizes the error maps between the restored results and the ground truth, where brighter colors indicate larger errors. As illustrated, our method obviously predicts depth maps with sharper edges, higher fidelity, and minimal errors. However, most existing methods struggle to generate precise boundaries and often suffer from various artifacts. For instance, as observed in Fig. 5, while almost all methods obtain precise edges, certain ones (e.g., DCTNet) still introduce redundant texture-copy artifacts. In contrast, our method produces results with both the fewest artifacts and minimal errors.

Additionally, the visual results further prove that our depth predictions exhibit superior structural preservation and the closest alignment with the ground truth. The visual results clearly demonstrate that our LGM-Net leverages complementary information from RGB guidance without transferring undesired textures, thereby elevating the visual quality of the predicted HR depth maps.

**Real-world Evaluation.** Table III presents the quantitative comparison results on the real-world RGB-D-D dataset. Following standard evaluation protocols, we first evaluate the models pre-trained on the NYU-v2 dataset directly on RGB-D-D to assess cross-domain generalization, and subsequently assess the models after retraining them specifically on this real-world dataset. As shown in the table, our LGM-Net consistently achieves state-of-the-art performance under both settings. Furthermore, compared to synthetic data, low-resolution depth maps captured in real-world environments often exhibit severe blurriness and structural distortions, rendering the reconstruction task considerably more challenging.

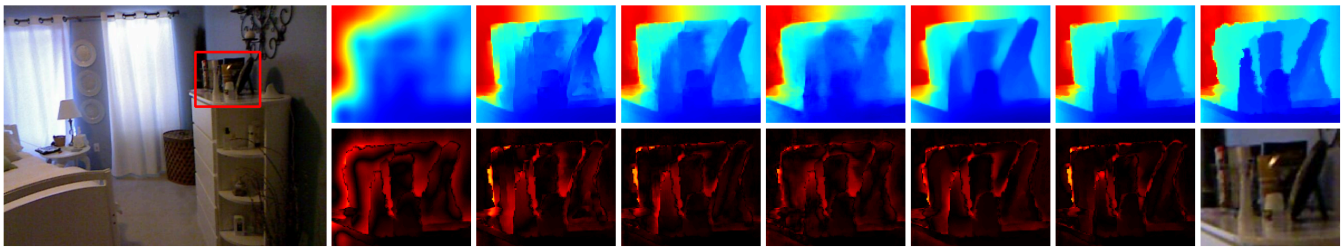
Fig. 8 provides visual comparisons on the RGB-D-D dataset. It can be observed that existing methods, such as DCTNet and SGNet, struggle to reconstruct sharp boundaries and frequently introduce blurring artifacts along complex object contours. In contrast, our approach recovers highly accurate depth structures with crisp edges. In summary, this quantitative and visual evidence demonstrates that our method effectively enhances depth structures and exhibits robust generalization in real-world applications.

**Joint GDSR and denoising.** Depth maps acquired in real-world environments are frequently corrupted by noise, which poses a substantial challenge for high-resolution depth restoration. Following the approach in [57], we inject Gaussian noise with a variance of 25 into the low-resolution depth inputs to simulate this degradation. Notably, owing to our proposed



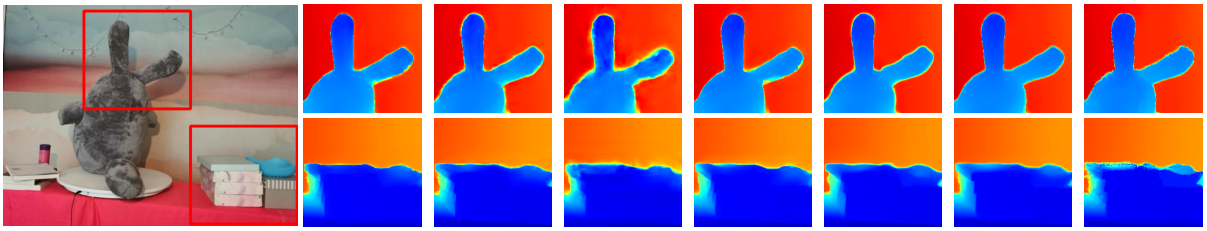
(a) RGB (b) Bicubic (c) DKN (d) JIIF (e) DCTNet (f) SGNet (g) Ours (h) GT & RGB

Fig. 6. Visual results and error maps on Middlebury dataset (×16).



(a) RGB (b) Bicubic (c) DKN (d) JIIF (e) DCTNet (f) SGNet (g) Ours (h) GT & RGB

Fig. 7. Visual results and error maps on NYU-v2 dataset (×16).



(a) RGB (b) DKN (c) FDSR (d) DCTNet (e) SUFT (f) SGNet (g) Ours (h) GT

Fig. 8. Visual results on real-world RGB-D-D dataset.

TABLE III
QUANTITATIVE COMPARISONS (RMSE) ON REAL-WORLD RGB-D-D DATASET.

| Methods | Training Dataset(s) | RMSE |
|---|---|---|
| DKN [48] | NYU-v2 / RGB-D-D | 7.38 / 5.44 |
| FDSR [43] | NYU-v2 / RGB-D-D | 7.50 / 5.49 |
| DCTNet [49] | NYU-v2 / RGB-D-D | 7.37 / 5.43 |
| SUFT [50] | NYU-v2 / RGB-D-D | 7.22 / 5.41 |
| SSDNet [52] | NYU-v2 / RGB-D-D | 7.32 / 5.38 |
| SGNet [56] | NYU-v2 / RGB-D-D | 7.22 / 5.32 |
| **Ours** | NYU-v2 / RGB-D-D | **7.14 / 4.66** |

training data augmentation pipeline, the model does not require retraining on these noisy datasets specifically; instead, we simply incorporate depth maps corrupted by random noise as part of the training set during the initial training stage. Table IV demonstrates that our LGM-Net outperforms competing methods on joint GDSR and denoising tasks across the NYU-v2 and Middlebury datasets. Compared with the second-best method (DAGF), our LGM-Net achieves a significant RMSE reduction of 0.45 (×8) and 0.70 (×16) on the NYU-v2 dataset, and 0.09 (×8) and 0.25 (×16) on the Middlebury dataset. These results convincingly validate that our LGM-Net exhibits superior robustness and generalization capability when handling noisy depth observations.

**3D reconstruction.** To further evaluate the advantage of our method in terms of 3D geometric structure, we project the predicted 2D HR depth maps into 3D space to visualize the reconstructed point clouds. As illustrated in Fig. 9, the visualizations show that existing methods often produce distorted boundaries and uneven artifacts on flat surfaces due to inaccurate depth inference, particularly around the complex edges of the cabinet. By comparison, our method reconstructs sharp object shapes while maintaining smooth planar structures. The visualization results from the 3D perspective solidly verify the capability of our LGM-Net to retain fine-grained geometric topologies in 3D perception tasks.

**Model efficiency.** Fig. 10 shows the efficiency map of different GDSR methods. The horizontal axis represents the number of parameters (annotated in parentheses), and the

TABLE IV
QUANTITATIVE COMPARISONS (RMSE) OF JOINT GDSR AND DENOISING TASKS ON NYU-V2 AND MIDDLEBURY DATASETS.

| Scale | DJF [6] | DJFR [46] | DSRNet [58] | PAC [59] | DKN [27] | DAGF [57] | **Ours** |
|---|---|---|---|---|---|---|---|
| NYU-v2 | | | | | | | |
| ×4 | 3.74 | 4.01 | 4.36 | 4.23 | 3.39 | 3.25 | **2.86** |
| ×8 | 5.95 | 6.21 | 6.31 | 6.24 | 5.24 | 5.01 | **4.56** |
| ×16 | 9.61 | 9.90 | 9.75 | 9.54 | 8.41 | 7.54 | **6.84** |
| Middlebury | | | | | | | |
| ×4 | 1.80 | 1.86 | 1.84 | 1.81 | 1.76 | 1.72 | **1.64** |
| ×8 | 2.99 | 3.07 | 2.99 | 2.94 | 2.68 | 2.61 | **2.52** |
| ×16 | 5.16 | 5.27 | 4.70 | 5.08 | 4.55 | 4.24 | **3.99** |

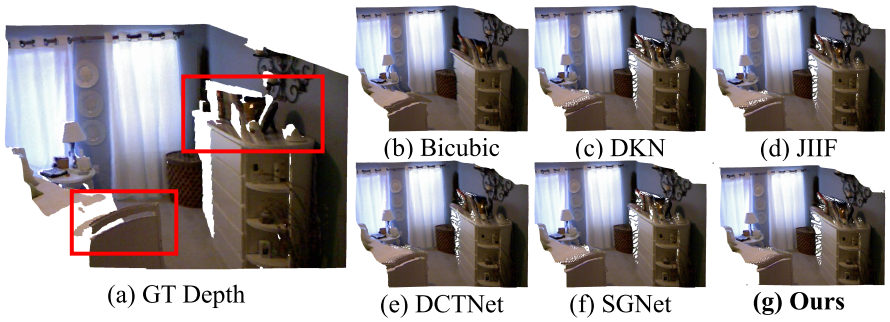



Fig. 9. Visual comparison of 3D point clouds reconstructed from the predicted depth maps.

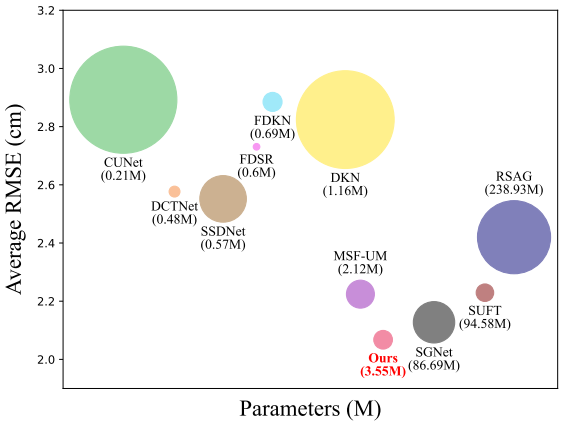



Fig. 10. Efficiency map of different GDSR methods. Smaller circle indicates faster inference speed.

vertical axis displays the average RMSE across 12 GDSR tasks (four benchmarks at three scales). Additionally, the circle area corresponds to the inference time (all tested on the same device for fair comparison), where smaller circles indicate faster speed. The efficiency map reveals that our method achieves state-of-the-art performance with relatively few parameters while maintaining fast inference speed. For example, although FDSR requires fewer parameters, its depth reconstruction accuracy significantly lags behind ours. Similarly, SGNet achieves the second-best average RMSE result, but it exhibits substantially more parameters and slower inference speed. In summary, our LGM-Net strikes an optimal balance among prediction accuracy, model capacity, and computational efficiency, highlighting its immense potential for real-time applications on edge devices.

### C. Ablation Study

We conduct ablation studies to validate the effectiveness of several components in our LGM-Net, including the SAGSP block, the three key modules (DCMA, MDAF, EADE), and the three loss functions ($L_{std}$, $L_{grd}$, $L_{esc}$). The quantitative results of these ablation studies are presented in Table V.

**Basic Block: SAGSP *vs.* Restormer.** The SAGSP block serves as the fundamental unit for efficient feature extraction, developed as an optimized variant of the Restormer block. To validate the efficacy of this block, we substitute it with the original Restormer block while maintaining nearly the same number of parameters. As shown in Table V, with nearly the same number of parameters, substituting our optimized SAGSP block (Model 1) for the original Restormer block (Model 2) reduces the average RMSE from 3.92 to 3.80. This demonstrates its efficient feature extraction capability under equivalent capacity constraints. Such advantage holds consistently even when all key modules are equipped (Model 9 *vs.* Final Model), yielding an additional 0.08 RMSE decrease. This indicates SAGSP block delivers superior feature extraction performance under equivalent parameter constraints.

**Effectiveness of DCMA.** The DCMA module is designed to promote bidirectional and selective interaction between RGB features and depth features. To demonstrate the function of this module, we disable it in Model 1, Model 4, Module 5 and Model 6, which means the output of each SAGSP block is passed directly to the subsequent stage without any cross-modal interaction. We conduct comparative analyses across four sets: Model 1 & Model 3, Model 5 & Model 7, Model

TABLE V
QUANTITATIVE RESULTS OF THE ABLATION STUDY ON INDIVIDUAL COMPONENTS FOR $\times 16$ GDSR.

| Model | Basic Block | | Key Modules | | | Loss Functions | | | | RMSE ($\times 16$ GDSR) | | | | Average |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | SAGSP | Restormer | DCMA | MDAF | EADE | $\mathcal{L}_1$ | $\mathcal{L}_{std}$ | $\mathcal{L}_{grd}$ | $\mathcal{L}_{esc}$ | Lu | Middlebury | RGB-D-D | NYU-v2 | |
| Model 1 | ✓ | | | | | ✓ | ✓ | ✓ | ✓ | 3.87 | 3.28 | 2.86 | 5.19 | 3.80 |
| Model 2 | | ✓ | | | | ✓ | ✓ | ✓ | ✓ | 4.03 | 3.37 | 2.95 | 5.32 | 3.92 |
| Model 3 | ✓ | | ✓ | | | ✓ | ✓ | ✓ | ✓ | 3.62 | 3.06 | 2.63 | 4.89 | 3.63 |
| Model 4 | ✓ | | | ✓ | | ✓ | ✓ | ✓ | ✓ | 3.78 | 3.21 | 2.80 | 5.15 | 3.70 |
| Model 5 | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | 3.70 | 3.16 | 2.73 | 5.04 | 3.66 |
| Model 6 | ✓ | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 3.67 | 3.13 | 2.71 | 4.94 | 3.55 |
| Model 7 | ✓ | | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | 3.53 | 2.96 | 2.53 | 4.75 | 3.44 |
| Model 8 | ✓ | | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | 3.61 | 3.03 | 2.60 | 4.85 | 3.52 |
| Model 9 | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 3.52 | 2.97 | 2.45 | 4.76 | 3.42 |
| Model 10 | ✓ | | ✓ | ✓ | ✓ | ✓ | | | | 3.64 | 3.05 | 2.51 | 4.83 | 3.51 |
| Model 11 | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | | | 3.60 | 3.01 | 2.48 | 4.80 | 3.47 |
| Model 12 | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | 3.54 | 2.96 | 2.43 | 4.74 | 3.41 |
| Final model | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | **3.46** | **2.89** | **2.37** | **4.63** | **3.34** |

4 & Model 8, and Model 6 & Final Model, we observe that the application of DCMA module substantially enhances the reconstruction quality, bringing remarkable RMSE reductions of 0.25, 0.22, 0.18, and 0.21. This significant promotion robustly proves the necessity of bidirectional cross-modal interaction carried out by DCMA module.

**Effectiveness of MDAF.** The MDAF module is implemented to achieve deep and comprehensive fusion of multi-modal features. To verify the contribution of this module, we replace it with a simple channel-wise concatenation operation in Model 1, Model 3, Model 5, and Model 7. Therefore, in these variants, the RGB and depth features are merely concatenated rather than adaptively fused. By comparing the following four sets including Model 1 & Model 4, Model 5 & Model 6, Model 3 & Model 8, and Model 7 & Final Model, we draw a conclusion that the activation of MDAF module indeed decreases the average RMSE by 0.10, 0.11, 0.03, and 0.10, thereby validating its crucial role in terms of comprehensive feature integration.

**Effectiveness of EADE.** The EADE module acts as a dynamic refiner to sharpen edge details in the predicted depth maps. To assess the effectiveness of this module, we remove it from Model 1, Model 3, Model 4, and Model 8, leaving the edge boundaries of predicted depth maps unenhanced. Comparative analyses of the four sets (Model 1 & Model 5, Model 4 & Model 6, Model 3 & Model 7, and Model 8 & Final Model) clearly prove that the models equipped with EADE module achieve apparently superior performance. Among these four comparison sets, EADE module improves the average RMSE values by 0.14, 0.15, 0.11, and 0.18, which definitely confirms its necessity for precise edge restoration.

**Impact of Loss Functions.** To explore the contribution of our auxiliary losses, we train the full model under different loss configurations. Starting from the $\mathcal{L}_1$-only baseline (Model 10, RMSE 3.51), adding the relative-scale loss $\mathcal{L}_{std}$ (Model 11) and gradient loss $\mathcal{L}_{grd}$ (Model 12) progressively lowers the RMSE to 3.47 and 3.41, respectively. Notably, when all loss functions are deployed together, our model achieves the best performance. The deployment of full supervision with the edge consistency loss $\mathcal{L}_{esc}$ (Final Model) achieves the best RMSE of 3.34. The comparison results validate the reasonability and effectiveness of our proposed joint loss constraints that effectively guide the network in terms of relative scale, gradient domain, and edge consistency.

## V. CONCLUSION

We propose LGM-Net, a lightweight and generalizable framework for multi-scale GDSR. By integrating cross-modal feature interaction, structural detail refinement, and unified cross-dataset training, our method addresses the core challenges of cross-scene generalization. Through architectural reconstruction and training strategy optimization, a single and once-trained model successfully accomplishes multi-scale GDSR (e.g., $\times 4$, $\times 8$, $\times 16$), achieving state-of-the-art performance with immense practical potential. Despite these advances, LGM-Net still exhibits two major limitations: synthetic degradations fall short of complex real-world distortions, and attention-based mechanism incurs computational overhead. To tackle these issues, our future work will focus on unsupervised domain adaptation to bridge the synthetic-to-real gap, as well as model compression techniques to facilitate real-time deployment on low-power edge devices.

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